ROADsense_AI

PROJECT PROPOSAL

Road Accident Analytics for Safer Roads

1. TITLE

RoadSense_AI – Road Accident Analytics for Safer Roads

2. PROBLEM STATEMENT

Road accidents are a major safety concern, and it can be difficult to identify which states and districts record higher numbers of accidents, injuries, and deaths. Raw accident data is not always easy to compare or interpret. Without clear analysis and visualization, users may struggle to understand where the recorded accident burden is higher and where further road-safety attention may be needed.

3. SOLUTION

RoadSense_AI will analyze the available 2022 road accident dataset at State and District level. The project will organize the data, prepare it for analysis, compare accident cases, injuries, and deaths across locations, and present findings through understandable charts and an interactive dashboard. The analysis will describe patterns in the available data; it will not claim that a road or route is guaranteed to be safe.

4. TARGET USERS

- Drivers and commuters
- Road travelers
- Traffic police and road-safety authorities
- Government and transport departments
- Researchers and students

5. QUESTIONS / QUERIES TO BE ANSWERED FOR THE USER

Question / Query	Variables Used	Expected Analysis
Which states and districts record a higher number of road accidents?	State/UT, District, Accident Cases	Compare and rank recorded accident cases by location.
Which locations record a higher number of injured people?	State/UT, District, Injured	Compare injured counts across locations.
Which locations record a higher number of deaths?	State/UT, District, Died	Compare death counts across locations.
Which locations show greater recorded accident impact?	State/UT, District, Accident Cases, Injured, Died	Compare accident, injury, and death measures together.

6. EXPECTED DASHBOARD

- Summary cards showing total recorded Accident Cases, Injured, and Died in the dataset.
- State-wise comparison of recorded accident cases.
- District-wise comparison of recorded accident cases.
- Lists of states and districts with higher recorded accident counts.
- Charts comparing accidents, injuries, and deaths.

- Map or location visualization where suitable location information is available.
- State and District filters where supported by the dataset.
- Short road-safety insights based on the analyzed records.

7. DATA VARIABLES

Variable	Type	Purpose
Year	Numerical	Identifies the reporting year (2022).
State/UT	Categorical	Identifies the state or location represented in the data.
District	Categorical	Identifies the district/location being analyzed.
Accident Cases	Numerical	Measures recorded road accident cases.
Injured	Numerical	Measures recorded number of people injured.
Died	Numerical	Measures recorded number of deaths.
Data Status	Categorical	Indicates whether a working-dataset value is official or estimated, where this field is available.
Source	Text	Records the source or source type for the row, where available.

8. EXPECTED OUTCOME

RoadSense_AI is expected to turn the available 2022 road accident dataset into understandable location-based insights. Users will be able to compare recorded accident cases, injuries, and deaths across states and districts using tables, charts, and a dashboard. Any estimated district-level values in the working dataset must be clearly distinguished from official figures, and conclusions will be limited to what the dataset supports.